

VOIS AICTE INTERNSHIP • MAJOR PROJECT SUBMISSION

Seasonal Agriculture Performance Analysis

A Data-Driven Study on Crop Yield, Water Efficiency & Farm Economics

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Program Track:	VOIS AICTE Batch 1 (2026–2027) • Data Visualization

Problem Statement

Core challenges addressed by the seasonal agriculture study

- **Seasonal Sensitivity:** Indian agricultural production is intrinsically tied to shifting climatic regimes across Kharif (Monsoon), Rabi (Winter), and Zaid (Summer). Shifts in rainfall distribution and temperature swings directly alter vegetative output and crop survival rates.
- **Data Opacity in Agriculture:** Raw farm-level agronomic records fail to explain the direct correlation between input investments (chemical fertilizers, pesticide dosages, water pumping) and realized harvest revenue, leaving farmers vulnerable to unhedged losses.
- **Resource Inefficiency & Waste:** Conventional practices, such as uncontrolled flood irrigation and blanket chemical application, create severe water waste and soil degradation without proportional yield increases.
- **Project Objective:** Leverage advanced Exploratory Data Analysis (EDA) and visualization on 4,000 multi-regional farm profiles to identify seasonal production disparities, benchmark irrigation technologies, and deliver data-backed recommendations for resilient farm planning.

Project Description & Analytical Architecture

- **Context & Mandate:** Capstone project executed under the VOIS AICTE Internship Batch 1 (2026–2027) within the Data Visualization specialization.
- **Dataset Scope & Diversity:** Comprehensive analysis of 4,000 empirical farm records spanning 8 key agricultural states (Andhra Pradesh, Maharashtra, Telangana, Karnataka, Gujarat, Tamil Nadu, Punjab, Madhya Pradesh) across 28 distinct agronomic, meteorological, and economic features.
- **Crops & Seasons Covered:** 8 major commercial and staple crops (Sugarcane, Chilli, Cotton, Groundnut, Maize, Pulses, Rice, Wheat) tracked across Kharif, Rabi, and Zaid seasons.
- **End-to-End Analytical Architecture:** 1) Median Imputation & Data Cleaning → 2) Univariate & Bivariate EDA → 3) Water Efficiency Benchmarking → 4) Crop Economics & Profitability Assessment → 5) Micro-Climate Pest Risk Modeling → 6) Strategic Policy Advisory.

Who Are the End Users?

Actionable value delivered across primary agricultural stakeholders

1. Farmers & Cooperatives

Provides clear empirical guidance for crop selection tailored to season-specific water availability, proving why switching from flood to drip irrigation increases net profits by over ₹1.4 Lakhs/ha.

2. KVK Extension Officers

Arms agricultural extension workers with quantitative risk thresholds (>65% relative humidity in Kharif) to issue timely preventive spraying advisories rather than reactive measures.

3. Policy Makers & Ministries

Enables state agricultural ministries to design evidence-based micro-irrigation capital subsidy schemes, seasonal credit allocations, and targeted Minimum Support Price (MSP) revisions.

4. Agri-Supply Chain Managers

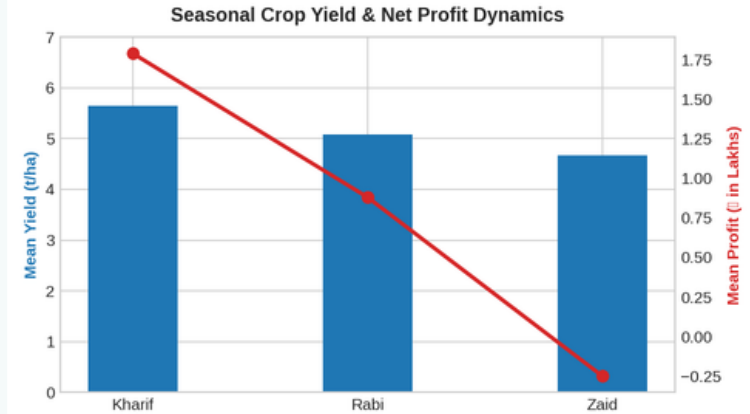
Assists storage operators and logistics providers in forecasting seasonal harvest surges, optimizing cold storage allocation, and preventing post-harvest supply gluts.

Technology Stack & Tools Used

- **Python (v3.10+):** Core programming language used for high-performance numerical computation, data wrangling, and workflow automation.
- **Pandas & NumPy:** Employed for tabular manipulation, handling missing records via seasonal median imputation, multi-group aggregations, and feature extraction across 4,000 records.
- **Matplotlib & Seaborn:** Leveraged to construct publication-grade visualizations: dual-axis comparative charts, grouped bar plots, scatter plots with seasonal hue, and correlation heatmaps.
- **Jupyter Notebook & VS Code:** Interactive scientific development environment for iterative exploratory data analysis (EDA), code experimentation, and visualization debugging.
- **Git & GitHub:** Version control and public repository hosting for open reproducibility and academic presentation.

RESULTS: Seasonal Yield & Profitability Dynamics

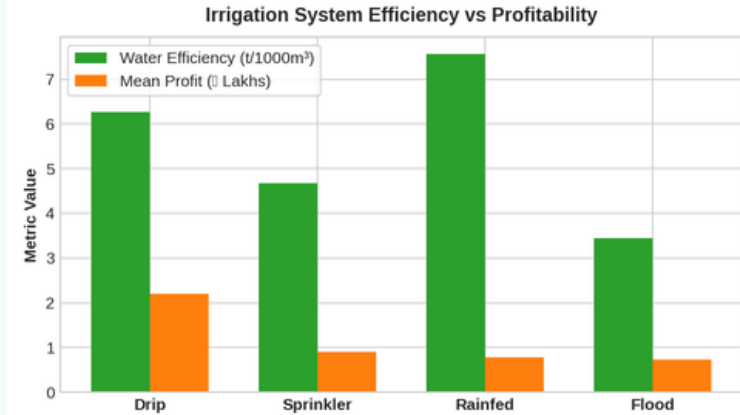
```
# 1. Seasonal Yield & Profit Dynamics
import pandas as pd, matplotlib.pyplot as plt
df = pd.read_csv('dataset.csv')
season = df.groupby('Season')[['Yield_Tonnes_Ha', 'Profit_INR']].mean()
season.reindex(['Kharif', 'Rabi', 'Zaid'])
```



- **Kharif Dominance:** Kharif generates the highest average yield (5.64 t/ha) and highest profit (₹1.79 Lakhs) supported by abundant monsoon rainfall (mean 852 mm).
- **Rabi Consistency:** Rabi yields an average 5.08 t/ha with stable returns (₹0.88 Lakhs), benefiting from lower disease pressure despite lower precipitation (436 mm).
- **Zaid Vulnerability:** Zaid experiences negative average profit (–₹0.25 Lakhs) and lowest yields (4.67 t/ha) due to extreme heat (31.0°C) and intense irrigation expenses.

RESULTS: Irrigation Method Efficiency & Economics

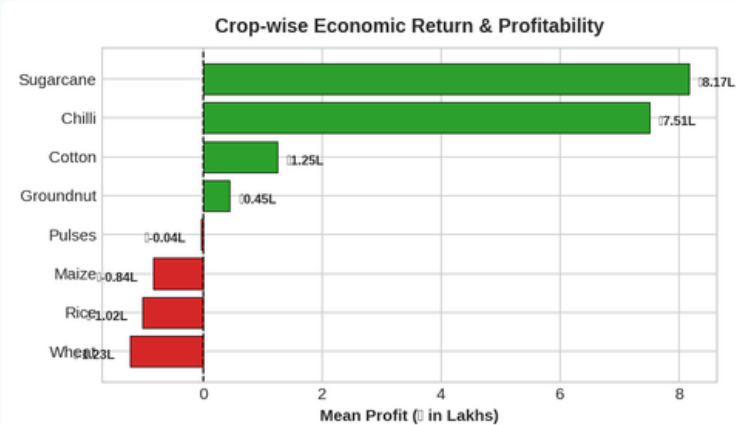
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# 2. Irrigation Efficiency
irr = df.groupby('Irrigation_Method')[
    ['Water_Efficiency_t_per_1000m3', 'Profit_INR']
].mean().reindex(['Drip', 'Sprinkler', 'Rainfed', 'Flood'])
```



- **Drip Superiority:** Drip irrigation achieves an efficiency of 6.27 t/1000m³ and highest mean profit of ₹2.20 Lakhs, drastically outperforming flood methods.
- **Flood Inefficiency:** Flood irrigation consumes an average of 8,026 m³ water per farm but results in low water efficiency (3.44 t/1000m³) and meager profits (₹0.73 Lakhs).
- **Economic Takeaway:** Adopting micro-irrigation increases net income by nearly 200% compared to traditional flood methods while cutting water consumption by 26%.

RESULTS: Crop-wise Profitability & Price Dynamics

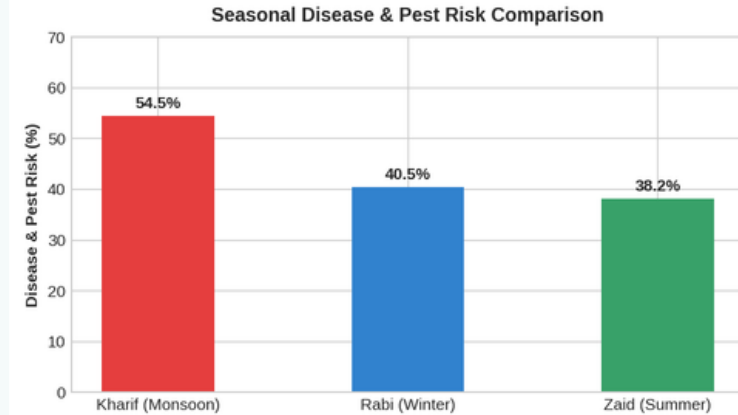
```
# 3. Crop Profitability
crop_perf = df.groupby('Crop')[
    'Profit_INR'
].mean().sort_values(ascending=False) / 1e5
```



- **High-Value Cash Crops:** Sugarcane (avg profit ₹8.17 Lakhs; yield 46.9 t/ha) and Chilli (avg profit ₹7.51 Lakhs; price ₹1,03,372/t) dominate farm profitability.
- **Staple Grains Margin Squeeze:** Traditional food grains suffer from negative margins (Rice: -₹1.02L, Wheat: -₹1.23L, Maize: -₹0.84L) when input and land costs outpace MSP rates.
- **Strategic Implication:** Farmers must be incentivized to practice inter-cropping and cash crop diversification to hedge against staple crop operational losses.

RESULTS: Humidity vs Disease & Pest Risk Analysis

```
# 4. Pest Risk vs Humidity
sns.scatterplot(
    data=df, x='Humidity_pct',
    y='Disease_Pest_Risk_pct', hue='Season'
)
```

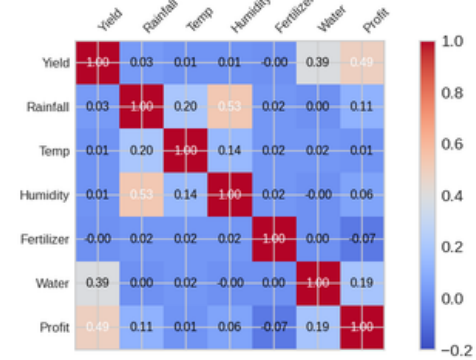


- **Monsoon Pest Spikes:** Kharif exhibits the highest disease/pest risk (mean 54.5%) strongly driven by high relative humidity (>70%) and warm ambient temperatures.
- **Dry Season Protection:** Rabi (40.5% risk) and Zaid (38.2% risk) present significantly lower biological threats due to lower atmospheric moisture.
- **Preventive Guidance:** KVK advisories should mandate bio-fungicide and pest monitoring as soon as ambient relative humidity crosses 65% in monsoon zones.

RESULTS: Agronomic & Financial Correlation Matrix

```
# 5. Correlation Matrix  
cols = ['Yield_Tonnes_Ha', 'Rainfall_mm',  
        'Avg_Temperature_C', 'Profit_INR']  
corr = df[cols].corr()
```

Correlation Matrix of Agronomic Indicators



- **Core Yield Drivers:** Crop Yield demonstrates a moderate positive correlation with net profit ($r = 0.49$) and water utilized ($r = 0.39$), highlighting input dependencies.
- **Over-Fertilization Penalty:** Fertilizer dosage shows zero correlation with yield ($r = -0.00$) and negative correlation with profit ($r = -0.07$), exposing chemical wastage.
- **Climatic Interdependence:** Rainfall, humidity, and soil moisture exhibit strong mutual correlation ($r = 0.45$ to 0.53), confirming clustered seasonal patterns.

Future Scope & Strategic Recommendations

- **Machine Learning Predictive Modeling:** Deploy regression and ensemble algorithms (Random Forest, XGBoost) to forecast crop yield and market revenue 60 days before harvest based on early seasonal climatic features.
- **Real-Time IoT & Soil Telemetry:** Integrate live field sensor telemetry (measuring N-P-K, soil moisture, and leaf wetness) to trigger automated precision drip irrigation and variable-rate fertilization.
- **Mobile Agricultural Advisory Dashboard:** Develop a lightweight multilingual mobile application (Streamlit / React) to deliver hyperlocal sowing advice and pest alerts to smallholder farmers in regional languages.
- **Policy-Driven Subsidies for Micro-Irrigation:** Use empirical water-efficiency benchmarks from this study to assist state governments in subsidizing Drip/Sprinkler installations in water-stressed districts.

Project GitHub Repository

- **Repository URL:** <https://github.com/DevSatyam7/Seasonal-Agriculture-Performance-Analysis>

Contains full Jupyter Notebooks, raw & processed datasets, visualization pipelines, and project documentation.

- **notebooks/Seasonal_Agriculture_Analysis.ipynb:** Complete Python Jupyter Notebook containing data cleaning, exploratory analysis, visualizations, and statistical summaries.
- **dataset/seasonal_agriculture_performance_dataset.csv:** Complete 4,000-row farm dataset with all 28 agronomic and financial features.
- **visualizations/:** Full-resolution PNG exports of all generated charts (Yield dual-axis, irrigation efficiency, crop profitability, correlation matrix).
- **README.md:** Executive summary, installation instructions, analytical findings, and reproduction guide.



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Certificate of Completion

Presented to

Satyam Kumar

For the successful completion of

Data Visualization

Issued on September 6, 2026

ID: VFLMS26_162920



Thank You!

Questions & Discussion

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